

Penalty and Barrier Methods (Luenberger/Ye, §B.1)

Consider constrained optimization

$$(*) \quad \min_{x \in S} f(x) \quad f: \mathbb{R}^n \rightarrow \mathbb{R}, \quad S \subset \mathbb{R}^n$$

constraint

Let's look at algorithms to (approximately) solve such problems by replacing them with unconstrained problem.

Penalty methods: Instead of (*) consider

$$\min_{x \in \mathbb{R}^n} f(x) + c P(x)$$

$c \dots$ penalty parameter
 $c > 0$

$P(x) \dots$ penalty function
for constraint $x \in S$

$P(x)$ satisfies: continuous in $\mathbb{R}^n$, $P(x) \geq 0$ and $P(x) = 0$ for $x \in S$.

Example:

$$S = \{x: g_i(x) \leq 0, i = 1, \dots, p\}, \text{ then we}$$

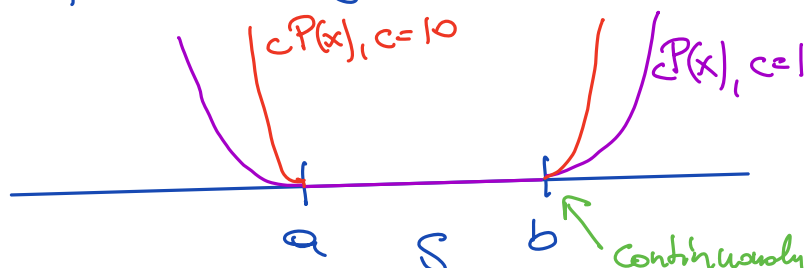
can use:

$$P(x) := \frac{1}{2} \sum_{i=1}^p [\max(0, g_i(x))]^2$$

for instance:

$$S = \{x \in \mathbb{R} \mid a \leq x \leq b\} \rightarrow \begin{aligned} a - x &\leq 0 \\ x - b &\leq 0 \end{aligned}$$

$$P(x) = \frac{1}{2} \max(0, a-x)^2 + \frac{1}{2} \max(0, x-b)^2$$

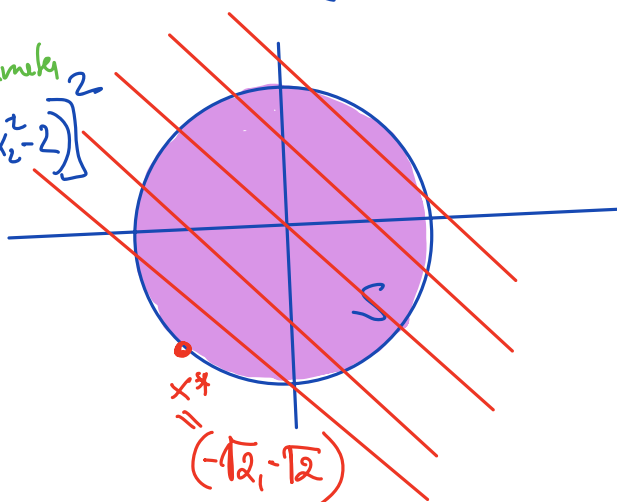


diff'able

Example: $\min_{(x_1, x_2) \in \mathbb{R}^2} x_1 + x_2$
 $(x_1, x_2) \in S$

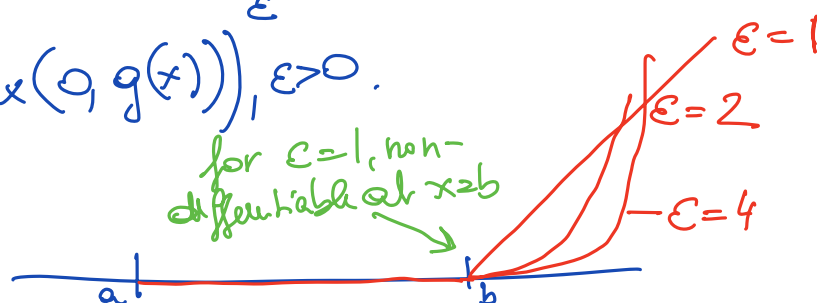
$$S = \{x \in \mathbb{R}^2 \mid x_1^2 + x_2^2 - 2 \leq 0\}$$

penalty formulation:
 $\min_{x \in \mathbb{R}} x_1 + x_2 + \frac{c}{2} [\max(0, q(x_1^2 + x_2^2 - 2))]^2$
 (penalty parameter)
 (penalty)



Other penalty functions:

$$P_2(x) = \left(\max(0, g(x)) \right)^{\frac{1}{\epsilon}}, \epsilon > 0.$$



In practice, we have to choose c large to approximately satisfy the constraint; we typically start with moderate c_k , solve the problem for solution x_k and then increase c_{k+1} :

$$0 < c_1 < c_2 < c_3 \dots \longrightarrow \infty$$

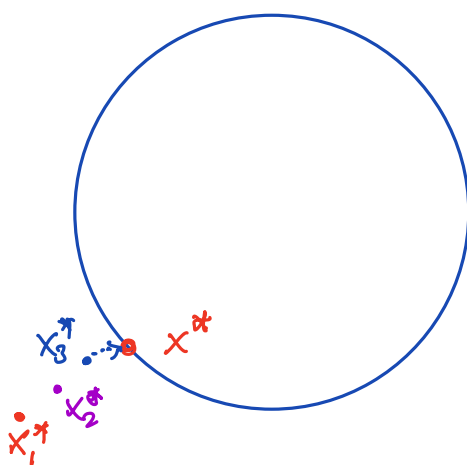
Solution to the problem

$$\min_{x \in \mathbb{R}^n} f(x) + c_k P(x)$$

is called x_k^* .

Theorem: The sequence $(x_k^*)_{k \geq 1}$ is produced by the penalty method. Then any limit is a solution to the constrained problem (*).

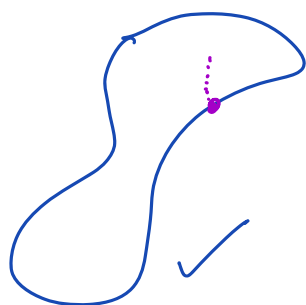
Why increase C_k slowly?
 Because the problems with increasing C_k become more difficult to solve.



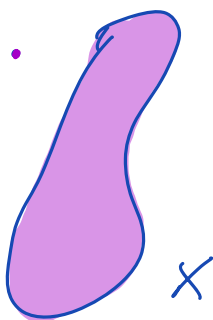
Barrier method

$$\min_{x \in S} f(x)$$

S is a "nice" set with non-empty interior and any point on the boundary of S can be approached from the interior.



cannot be approached from interior



no interior points!

Let:

$$S = \{x \in \mathbb{R}^n : g_i(x) \leq 0, i=1,2,\dots,p\}$$

Barrier methods establish a barrier at the boundary to prevent an algorithm from ever leaving the domain S .

Barrier function B satisfies: (i) continuous inside S

$$(ii) B(x) \geq 0$$

$$(iii) B(x) \rightarrow \infty \text{ as } x \text{ approaches the boundary}$$

Example 1: $B(x) = - \sum_{i=1}^p \frac{1}{g_i(x)}$ is a barrier function.

$B(x)$ continuous, $B(x) \geq 0$, x approaches boundary,

$$\text{then } g_i(x) \rightarrow 0 \rightarrow \frac{1}{g_i(x)} \rightarrow \infty.$$

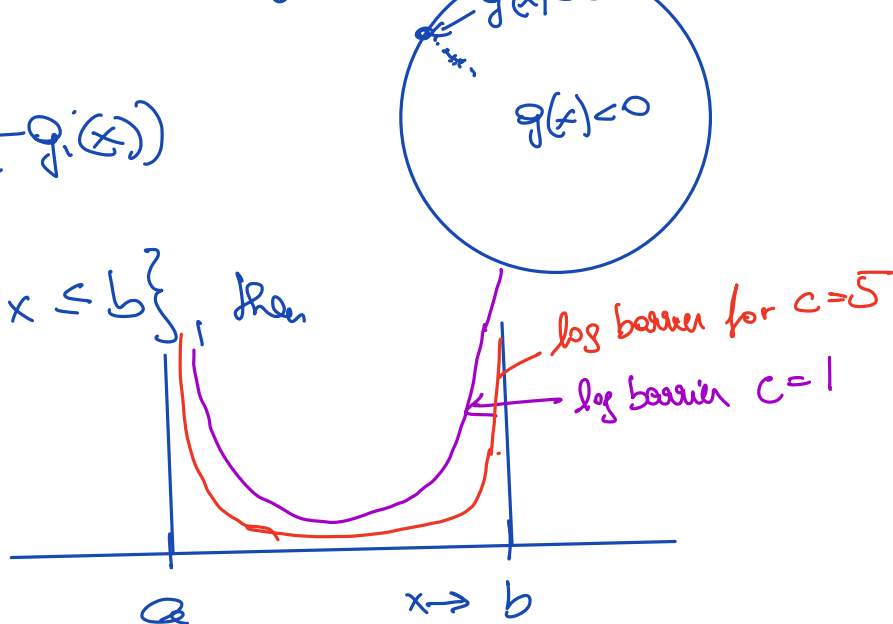
$$0 < 1 = 0$$

Example 2:

$$B(x) = - \sum_{i=1}^p \log(-g_i(x))$$

For $S = \{x \in \mathbb{R} : a \leq x \leq b\}$, then

B looks like:



Barrier method replaces constrained problem (*) with

$$\min_{\substack{x \in \mathbb{R}^n \\ x \text{ is interior of } S}} f(x) + \frac{1}{c} B(x)$$

for $c > 0$, $c \rightarrow \infty$ to approx (*)
can be solved with methods from unconstrained optimization (steepest descent, Newton, BFGS, ...)

alternative formulation
($\frac{1}{c} := \mu$)

$$\min_{\substack{x \in \mathbb{R}^n \\ x \text{ in interior}}} f(x) + \mu B(x)$$

$\mu > 0$ $\mu \rightarrow 0$ to approximate (*)

Again, let choose $c_1 < c_2 < \dots \rightarrow \infty$

[or equivalently, $\mu_1 > \mu_2 > \mu_3 \rightarrow 0$]

and denote by x_k^* the solution to

$$\min_{\substack{x \in \mathbb{R}^n \\ x \text{ in interior of } S}} f(x) + \frac{1}{c_k} B(x)$$

Then, any limit point of x_h^* is again a solution of the original constrained problem x^* .

In contrast to the penalty method, the barrier method approaches optimal points from the interior:

